

Barret Schloerke

Senior Software Engineer, Posit

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code[c("architect", "manager", "philosopher")]
data[c("wrangler", "explorer", "visualizer")]
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Summary

Senior software engineer on Shiny, Posit's web framework for R and Python (1M+ monthly downloads). 2,300+ merged pull requests across 160+ GitHub repositories since 2018. Current work: OpenTelemetry for Shiny, LLM-powered apps, and React-based custom UI. PhD in Statistics from Purdue University, focused on large data visualization.

Experience

Senior Software Engineer, Posit · Remote

Jan 2022 – Present

- Core developer of Shiny for Python (200K+ monthly downloads), a framework for interactive web apps.
- Cut CI time 98%, measured by 4+ hours of testing finishing in a 4.5-minute run, by architecting Playwright controllers for every Shiny for Python component and spreading 1000+ tests over 170 jobs (5 Pythons × 3 browsers).
- Built shinylive for R and Python, which export Shiny apps to static sites that run in the browser with WebAssembly.
- Enabled every Shiny user to trace apps with OpenTelemetry, build LLM-powered apps (querychat, shinychat), and pair agentic coding with React-based UI (shinyreact).

Software Engineer, RStudio (now Posit) · Remote

Jan 2018 – Dec 2021

- Core developer of Shiny, an R package for building interactive web apps in R.
- Cut release testing from 3 weeks across 10 people to under 2 days for 2 by architecting shinycoreci: nightly GitHub Actions testing of 150+ apps on Mac, Windows, and Linux × 5 R versions.
- Migrated Shiny's core JavaScript to TypeScript, giving the team confidence to change client code, by porting the client library to typed modules.
- Sustained plumber, reactlog, shinytest2, and dozens of other Shiny packages by owning releases, docs, and triage.

Tool Development Team, Gates Foundation: HBGDki · Remote

Apr 2016 – Dec 2018

- Produced custom summary statistics for 75+ datasets to feed visualization apps.
- Protected sensitive study data by coordinating secure data-handling practices with five other data scientists.
- Standardized the team's R package development by creating and maintaining template R packages.

Researcher, DARPA XDATA and D3M grants, Purdue University · West Lafayette, IN

Sep 2016 – Dec 2017

- Delivered a web interface for subject matter experts to explore machine learning pipelines by partnering remotely with Hafen Consulting, Kitware, and KnowledgeVis.
- Led service integration across 15 partner working groups by serving as the lead point of contact.

Software Engineer, Metamarkets · San Francisco, CA

Jan 2011 – Jul 2012

- Built interactive D3 visualizations on DVL, a reactive data visualization library, in CoffeeScript.
- Wrote Node.js front-end servers that automated deploys, clusters, and company / user management, allowing anyone on the team to ship or debug the client experience.
- Reduced site startup time and server load by trimming data queries and startup work.

Data Scientist Intern, Novartis Pharmaceutical R&D · Basel, Switzerland

May 2008 – Aug 2008

- Equipped management with a live view of employee availability by building a Gantt chart tool.
- Standardized common graphics for the R&D team by writing a web tutorial that packaged them as reusable plotting layers.

Education

PhD, Statistics, Purdue University · West Lafayette, IN

2013 – 2017

- Advisors: William Cleveland and Ryan Hafen. Emphasis: large data visualization.
- National Science Foundation Graduate Research Fellow (2012 – 2017).

MS, Mathematical Statistics, Purdue University · West Lafayette, IN

2013 – 2014

PhD program, Statistics, Rice University · Houston, TX

2012

- Advisor: Hadley Wickham.

- Statistics graphics research with Dianne Cook and Heike Hofmann.

Open source software

2,300+ merged pull requests across 160+ GitHub repositories since 2018. Packages I maintain or author, by merged PRs:

shiny Python · author · 350+ PRs · A web development framework for Python.

- Agent Skills (2026): Skills shipped in the package so coding agents know how to write Shiny apps.
- Test mode (2026): `test_server()` and exported test values for fast tests without a browser.
- Session lifecycle (2026): `session.destroy()`, a soft teardown when a session closes, and `session.allow_reconnect()`.
- Release train skill (2026): An Agent Skill that walks a py-shiny release from prep through PyPI, conda-forge, and shinylive.
- OpenTelemetry (2026): Traces and logs for sessions, reactivity, and outputs.
- Bookmarking (2025): Save and restore an app's state in the URL or on the server.
- Data frames (2024): `render.data_frame` with editing, selection, sorting, filtering, and styles.
- Output renderers (2024): `shiny.render.renderer`, a base class for writing your own output renderers.
- API reference (2023 – 2026): The quartodoc function reference and its runnable examples.
- UI components from bslib (2023 – 2024): Ported sidebars, cards, accordions, value boxes, tooltips, and popovers from R's bslib.
- Playwright testing (2023 – 2024): `shiny.playwright` controllers for every UI component, and the `shiny.pytest` plugin.

shinyreact R, Python, TypeScript · maintainer · 170+ PRs · Client-Side 'React' Interface for 'shiny'

- TypeScript: npm package (2026): A dual build published as `@posit-dev/shinyreact`, usable with no server-side JS.
- TypeScript: Wire protocol (2026): One versioned protocol, checked by a handshake, shared with the R and Python packages.
- Python: Agent Skills (2026): Skills for building and converting apps with coding agents, shipped in the package.
- R: Agent Skills (2026): Skills for building and converting apps with coding agents, shipped in the package.
- R: Initial development (2026): Ported the Python API to R, then CRAN checks, a pkgdown site, and webR builds for 0.1.0.
- Python: Renderer dependencies (2026): Find and serve the HTML dependencies of Shiny outputs used inside React.
- Python: ui.tsx apps (2026): Write the whole page in React and serve it from Python.
- R: ui.tsx apps (2026): Write the whole page in React and serve it from R.
- TypeScript: React hooks (2026): `useShinyInput` / `useShinyOutput` hooks, and `ShinyOutput` for classic Shiny outputs.
- Python: Testing (2026): Playwright e2e tests, WireTap for wire payloads, and in-memory `test_server()`.
- Python: Initial development (2026): Began as `shinyjson`, a JSON spec renderer, then turned React-first and dropped `json-render`.

GGally R · maintainer · 150+ PRs · Extension to 'ggplot2'

- S7 (2025): Moved `ggmatrix` to S7 for `ggplot2` 4.0.
- Discrete stats (2020): Count, cross, table, and statistic plots for `GGally` 2.0.
- `ggduo`, `ggnostic`, and friends (2016 – 2017): Plot matrices for two groups of variables, model diagnostics, and time series.
- `ggpairs` and `ggmatrix` (2014 – 2017): The generalized pairs plot and the plot matrix behind it.

shinytest2 R · maintainer · 140+ PRs · Testing for Shiny Applications

- Migration from `shinytest` (2022): `migrate_from_shinytest()` converts existing `shinytest` tests.
- Screenshots (2021 – 2023): Consistent screenshots, compared with a kernel density smoother to ignore stray pixels.
- Test recorder (2021 – 2022): `record_test()` writes a test while you click around the app.
- AppDriver (2021 – 2022): Drive a Shiny app in headless Chrome: set inputs, click, and snapshot values.

shiny R, TypeScript · author · 140+ PRs · Web Application Framework for R

- R: MCP Apps (2026): Serve Shiny apps as MCP Apps, with author tools for LLM clients.
- R: Session lifecycle (2026): `session$destroy()`, a soft teardown when a session closes, and clean state on reconnect.
- R: OpenTelemetry (2025): Traces and logs for sessions, reactivity, and outputs through `{otel}`.
- R: Quosures (2021): `exprToFunction()` and `installExprFunction()` accept quosures, including `inject()`ed ones.
- TypeScript: TypeScript (2021 – 2022): Ported `shiny.js` to TypeScript, moved the build to `esbuild`, and published the types.
- R: `devmode()` (2020 – 2021): One switch that turns on the options you want while developing an app.
- R: `testServer()` (2020): Test server functions and modules in memory against a mock session.
- R: Plot brushing (2018): Plot click and brush events carry CSS and image coordinates, and brushing works in WebKit.
- R: `reactlog` integration (2018 – 2020): Record every reactive event so `reactlog` can draw the reactive graph.

[plumber](#) R · maintainer · 130+ PRs · An API Generator for R

- `pr_*`() API (2020): The pipe-friendly router API and consistent R6 methods of plumber 1.0.
- Docker (2020 – 2021): Docker images for running plumber APIs.
- Async (2019 – 2020): Promise and future support for long-running endpoints.
- Serializers and parsers (2019 – 2020): Body parsers chosen by content type, and serializers for many output formats.
- OpenAPI docs (2018 – 2020): OpenAPI v3 specs and pluggable docs pages (Swagger, RapiDoc, Redoc).

[learnr](#) R · author · 70+ PRs · Interactive Tutorials for R

- Exercises (2019 – 2020): Safer exercise evaluation, checker functions, and clearer errors.
- Quiz questions (2019 – 2020): Questions rebuilt as Shiny modules, with a custom question API.

[promises](#) R · maintainer · 40+ PRs · Abstractions for Promise-Based Asynchronous Programming

[leaflet](#) R · author · 40+ PRs · Create Interactive Web Maps with the JavaScript 'Leaflet' Library

[reactlog](#) R · maintainer · 40+ PRs · Reactivity Visualizer for 'shiny'

[shinyloadtest](#) R · maintainer · 30+ PRs · Load Test Shiny Applications

[htmltools](#) R · author · 30+ PRs · Tools for HTML

[shinylive](#) R · maintainer · 20+ PRs · Run 'shiny' Applications in the Browser

[shinyswatch](#) Python · author · 20+ PRs · Bootswatch + Bootstrap 5 themes for Shiny.

[chromote](#) R · author · 17 PRs · Headless Chrome Web Browser Interface

[sortable](#) R · author · 12 PRs · Drag-and-Drop in 'shiny' Apps with 'SortableJS'

[trelliscopejs](#) R · author · 11 PRs · Create Interactive Trelliscope Displays

[lmmr](#) R · author · 9 PRs · Parallel Linear Mixed Model

[swagger](#) R · author · 9 PRs · Dynamically Generates Documentation from a 'Swagger' Compliant API

[leaflet.providers](#) R · maintainer · 8 PRs · Leaflet Providers

[autocogs](#) R · author · 8 PRs · Automatic Cognition Summaries

[shinychat](#) R, Python · author · 8 PRs · Chat UI Component for 'shiny'

[lazyrmd](#) R · author · 7 PRs · Render R Markdown Outputs Lazily

[leaflet.extras](#) R · author · 7 PRs · Extra Functionality for 'leaflet' Package

[gqlr](#) R · maintainer · 7 PRs · 'GraphQL' Server in R

[packagedocs](#) R · author · 6 PRs · Build Website of Package Documentation

[sass](#) R · author · 6 PRs · Syntactically Awesome Style Sheets ('Sass')

[querychat](#) R, Python · author · 5 PRs · Filter and Query Data Frames in 'shiny' Using an LLM Chat Interface

[leaflet.esri](#) R · author · 4 PRs · 'ESRI' Bindings for the 'leaflet' Package

[gt](#) R · author · 3 PRs · Easily Create Presentation-Ready Display Tables

Also contributed to: [bslib](#), [shinylive](#), [htmltools](#), [posit-sdk](#), [shinytest](#), [httpuv](#), [meetupr](#), [rmarkdown](#), [chatlas](#), [websocket](#), [flexdashboard](#), [webshot2](#), [cachem](#), [later](#), [vetiver](#), [testthat](#), [trelliscope](#), [webshot](#), [webdriver](#), [fastmap](#), [narwhals](#), [plumber2](#), [graphql](#), [ggplot2](#), [redoc](#), [shinyvalidate](#), [jquerylib](#), [routr](#), [brand.yml](#).

Earlier research software

rbokeh , Purdue University · R interface to the Bokeh visualization library.	2016
DVL , Metamarkets · Functionally reactive JavaScript library binding data to visualizations.	2011 – 2012
helpr , Rice University · Friendlier HTML documentation for R, with search across packages.	2010
cranvas , Iowa State University · Interactive plotting canvas for R, drawing a million points in under a second.	2010
tourr , Iowa State University · Geodesic interpolation and basis generation for tour methods in R.	2009
DescribeDisplay , Iowa State University · Publication-quality R graphics from GGobi plots.	2009
geozoo , Iowa State University · Library of high-dimensional geometric objects, including the new Hyper Ring Torus.	2007

Publications

Gökalp Yavuz, F., and **B. Schloerke** (2020). [Parallel computing in linear mixed models](#). *Computational Statistics* 35(3): 1273–1289.

Schloerke, B., H. Wickham, D. Cook, and H. Hofmann (2016). [Escape from Boxland: Generating a Library of High-Dimensional Geometric Shapes](#). *The R Journal* 8(2): 243–257.

Emerson, J., W. Green, **B. Schloerke**, J. Crowley, D. Cook, H. Hofmann, and H. Wickham (2013). [The Generalized Pairs Plot](#). *Journal of Computational and Graphical Statistics* 22(1).

Hofmann, H., D. Cook, C. Kielion, **B. Schloerke**, J. Hobbs, A. Loy, L. Mosley, D. Rockoff, Y. Huang, D. Wrolstad, and T. Yin (2011). [Delayed, Canceled, on Time, Boarding... Flying in the USA](#). *Journal of Computational and Graphical Statistics* 20(2).

Mosley, L., D. Cook, H. Hofmann, C. Kielion, and **B. Schloerke** (2010). [Monitoring the 2008 Election Visually](#). *Chance* 23(3).

Talks & workshops

Beyond Bootstrap: Building Custom Shiny UI with React · posit::conf(2026)	Sep 2026
OpenTelemetry in R: Capturing Traces, Logs, and Metrics for Shiny Applications · Open Source in Pharma	Apr 2026
{GGally}: Success and Long-Haul Maintenance Strategies · ggplot2 extenders · video	Nov 2025
Observability at scale: Monitoring Shiny Applications with OpenTelemetry · posit::conf(2025) · video	Sep 2025
{plumber2}: Streamlining Web API Development in R · useR! 2025 · video	Aug 2025
{querychat}: Natural language interfaces for interactive data graphics in Shiny · JSM 2025	Aug 2025
AI Kung-Fu : Training your LLM client in two function calls · ShinyConf 2025	Apr 2025
{shinylive}: Serverless Shiny applications workshop. An exercise in deploying your app to GitHub Pages · R/Pharma 2024 · video	Oct 2024
Data frames in Shiny for Python · posit::conf(2024) · video	Aug 2024
{shinylive}: Serverless Shiny apps · ShinyConf 2024 · video	Apr 2024
Lessons learned from testing 2500+ Shiny Apps every day · ShinyConf 2023 · video	Mar 2023
{shinytest2}: Regression testing Shiny applications · Johnson & Johnson Shiny Day	Oct 2022
{shinytest2}: Regression testing Shiny applications · rstudio::conf(2022) · video	Jul 2022
{shinytest2}: Regression Testing for Shiny Applications · R/Pharma 2022	May 2022
shinytest2: Regression Testing for Shiny Applications · ShinyConf 2022	Apr 2022
plumber + future: Async Web APIs · Harvard R User Group · video	Aug 2021
An intro to plumber · Harvard R User Group · video	Aug 2021
plumber + future: Async Web APIs · rstudio::global(2021) · video	Jan 2021
Expanding R Horizons: Integrating R with Plumber APIs · RStudio Webinar · video	Oct 2020
Episode 14: Shining a Light on learnr · Shiny Developer Series	Oct 2020
Episode 13: Inside Plumber 1.0 · Shiny Developer Series	Sep 2020
Episode 12: reactlog · Shiny Developer Series	Sep 2020
Reactlog 2.0: Debugging the State of Shiny · Statistical Programming DC	Apr 2020
Reproducible Shiny apps with shinymeta · ABACUS 2019	Nov 2019
Debug Large Data Apps: shiny + reactlog + sparklyr · Advanced R workshop, Northeastern University	May 2019
Reactlog 2.0: Debugging the State of Shiny · rstudio::conf(2019) · video	Jan 2019
Shiny · NASA Datanauts	Aug 2018
Cognostics: Metrics enabling detailed interactive visualization of big data · CSESC 2017	Apr 2017
Web Scraping with R	Mar 2017
tidyverse[c("magrittr", "dplyr", "tidyr")] · Purdue Graduate Statistics Seminar	Jan 2017
Cognostics: Metrics enabling detailed interactive visualization of big data · JSM 2016	Aug 2016
ggduo: Pairs plot for two group data · useR! 2016	Jun 2016
Analysis and Visualization of Large Complex Data with Tessera · Spring Research Conference, IIT	Jun 2016

Analysis and Visualization of Large Complex Data with Tessera · Purdue Graduate Statistics Seminar	Apr 2016
Analysis and Visualization of Large Complex Data with Tessera · Iowa State Graphics Research Group	Mar 2016
Analysis and Visualization of Large Complex Data with Tessera · NUMBAT Seminar, Monash University	Mar 2016
gqlr = GraphQL + R: a data query language and runtime · WOMBAT 2016	Feb 2016
Web Scraping with R · American Credit Acceptance	Oct 2015
Trelliscope: D&R visualization tool · Fields Institute Big Data Visualization	Feb 2015
ggplot2: displaying spatial and temporal data · Purdue Spatial Statistics Seminar	Oct 2014
git · Purdue Working Group	Jun 2014
Reducing Working Environment Inefficiencies · Purdue Graduate Statistics Seminar	Feb 2014
helpR: Help for R · Iowa State Working Group	Sep 2010
GGally: A Plot Matrix for All Variable Types · JSM 2010	Aug 2010

Videos

Git Worktree-First Workflow + Conductor	Aug 2026
Shiny 1.2 Release: Wrap your data frame with Narwhals!	Oct 2024
Getting Started with {shinytest2} Part I · Example + basics	Jul 2022
Getting Started with {shinytest2} Part 3 · Using shiny.testmode in {shinytest2}	Jul 2022
Getting Started with {shinytest2} Part 2 · Exporting values	Jul 2022
Programming Games with Shiny · Roll the Dice: with Quosures!	Jun 2022
Programming Games with Shiny · Roll the Dice	May 2022
Programming Games with Shiny · Guess the Number	May 2022
Programming Games with Shiny · Dragon Realm	Mar 2022
Maximize computing resources using future_promise()	Feb 2022
{reactlog} Rundown	Feb 2022

Blog posts

Introducing shinyreact: React UI backed by a Shiny server	Sep 2026
Shiny for Python 1.8	Sep 2026
Shiny updates: R 1.14, Python 1.7, bslib 0.12	Aug 2026
Bringing OpenTelemetry to R in production	May 2026
Shiny for Python 1.6 brings toolbars and OpenTelemetry	Apr 2026
OpenTelemetry + Shiny for R v1.12	Dec 2025
Tool Calling UI in shinychat	Nov 2025
promises v1.5.0	Nov 2025
Shiny at posit::conf(2025)	Sep 2025
Posit at SciPy 2025	Jul 2025
Shiny 1.4 brings bookmarking and Generative AI docs	Apr 2025
Shiny for Python 1.2.0	Oct 2024
Testing Shiny Apps in Python	Oct 2024
Announcing Shiny for Python 1.0	Jul 2024
Editable data tables in Shiny for Python 0.9.0	May 2024
Introducing: Shiny Express	Jan 2024
Shiny for Python 0.6.1	Dec 2023
Shiny for Python 0.6.0	Nov 2023
Shiny for Python 0.5.0	Aug 2023
Shiny for Python 0.4.0	Jun 2023
Shiny for Python out of alpha	Apr 2023

plumber 1.1.0	Mar 2021
Shiny 1.6	Feb 2021
learnr 0.10.0	Dec 2019
leaflet 2.0.0	May 2018

Honors & awards

National Science Foundation Graduate Research Fellowship · the only Computational Statistics awardee of 13 statistics recipients in 2012	2012 – 2017
Visiting Scientific Researcher, Monash University · large data exploration tools with NUMBAT	2016
Google Summer of Code, GGally::ggduo · pairs plots for two groups of variables	2016
Best Dynamic / Interactive Visualization, Data Insight SF	2011
NSF Graduate Research Fellowship, honorable mention	2011
ASA Data Expo, 2nd place (Iowa State working group)	2009

Teaching & mentoring

Internship Mentor , Posit · Samuel Bharti: a bioinformatics Shiny showcase , 10 Shiny apps built on 5 packages	2026
Workshop Teaching Assistant , rstudio::conf / posit::conf · Shiny in Production (2019), Applied Machine Learning (2020), Level Up with Shiny for R (2024), Shiny for R (2025)	2019 – 2025
Internship Mentor , RStudio · Leslie Huang: leaflet.providers and education materials	2019
Internship Advisor , RStudio · Daniel Chen: gradethis , a companion package for grading learnr exercises	2019
Instructor , R for Data Science Seminar , Purdue University	2017
Teaching Assistant , Big Data Analytics: Statistics and Data Visualization, Purdue University	2017
Teaching Assistant , Hadley Wickham's R Master Class, San Francisco and Chicago	2011 – 2015

Service

President , Statistics Graduate Student Office, Purdue University · Ran a town hall, a graduate student mentor program, and an esteemed speaker event with Arthur Dempster.	2016 – 2017
Graduate Student Mentor , Purdue University Statistics	2014 – 2016
StatCom Assistant , Purdue University	2014 – 2016

Skills

Languages	R, Python, TypeScript, JavaScript, HTML, CSS, SQL, Bash
Frameworks	Shiny (R and Python), React, plumber, htmltools, ggplot2, tidyverse
Testing & CI	Playwright, pytest, testthat, shinytest2, GitHub Actions
Publishing	CRAN, PyPI, npm, conda-forge, pkgdown, Quarto, GitHub Pages